King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 201 Final Exam Term 253

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. $\int_0^1 \int_0^2 (2x - y^2)\, dy\, dx =$

(a) $-\frac{1}{3}$

(b) 2

(c) $\frac{8}{3}$

(d) $\frac{4}{3}$

(e) $-\frac{2}{3}$

2. In cylindrical coordinates, $\int_0^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \int_{x^2+y^2}^9 y\, dz\, dy\, dx =$

(a) $\int_{-\pi/2}^{\pi/2} \int_{-3}^3 \int_{r^2}^9 (r^2 \sin\theta)\, dz\, dr\, d\theta$

(b) $\int_0^{\pi/2} \int_0^3 \int_{r^2}^9 (r^2 \sin\theta)\, dz\, dr\, d\theta$

(c) $\int_{-\pi/2}^{\pi/2} \int_0^3 \int_{r^2}^9 (r^2 \sin\theta)\, dz\, dr\, d\theta$

(d) $\int_0^{\pi/2} \int_0^3 \int_{r^2}^9 (r \sin\theta)\, dz\, dr\, d\theta$

(e) $\int_0^{\pi} \int_0^3 \int_r^3 (r \sin\theta)\, dz\, dr\, d\theta$

3. If $R = \{(x, y) : x^2 + y^2 \leq 4,\ x \leq 0\}$, then $\iint_R e^{-(x^2+y^2)/2}\, dA =$

(a) $(e^{-2} - 1)\frac{\pi}{2}$

(b) $(2 - e^{-2})\pi$

(c) $(1 - e^{-2})\frac{\pi}{2}$

(d) $(1 - e^{-2})\pi$

(e) $(1 - e^{-2})\frac{\pi}{4}$

4. The graph of the function $f(x, y) = 4 + x - y - x^2y + xy^2$ has

(a) one local maximum point and one local minimum point

(b) one saddle point and one local maximum point

(c) one saddle point and one local minimum point

(d) two local maximum points

(e) two saddle points

5. If $ax + by + cz + 1 = 0$ is an equation for the tangent plane to the surface $z = (x - y)^2$ at the point $(1, 2, 1)$, then $a + b + c =$

(a) 1

(b) -4

(c) 0

(d) -2

(e) 3

6. A set of parametric equations for the line that passes through the point $(4, 3, 2)$ and is perpendicular to both vectors $\vec{u} = \langle 2, 1, 1\rangle$ and $\vec{v} = \langle 1, 2, 1\rangle$ is given by

(a) $x = 4 - 2t,\ y = 3 - 2t,\ z = 2 + 2t,\ t \in (-\infty, \infty)$

(b) $x = 4 + t,\ y = 3 + t,\ z = 2 + 3t,\ t \in (-\infty, \infty)$

(c) $x = 4 + t,\ y = 3 + 2t,\ z = 2 - 2t,\ t \in (-\infty, \infty)$

(d) $x = 4 - t,\ y = 3 - t,\ z = 2 + 3t,\ t \in (-\infty, \infty)$

(e) $x = 4t,\ y = 3t,\ z = 2t,\ t \in (-\infty, \infty)$

7. The volume of the solid in the first octant bounded by the planes $z = 1$ and $3x + 3y + z = 4$ is equal to

(a) $\frac{3}{4}$

(b) 1

(c) $\frac{2}{3}$

(d) $\frac{1}{2}$

(e) $\frac{1}{3}$

8. The volume of the solid bounded above by the parabolic cylinder $z = 4 - x^2$ and below by the paraboloid $z = x^2 + 2y^2$ is equal to

(a) $\dfrac{\sqrt{2}\,\pi}{4}$

(b) 2π

(c) $\dfrac{3\pi}{2}$

(d) 4π

(e) $2\sqrt{2}\,\pi$

9. The limit $\lim_{(x,y)\to(0,0)} \dfrac{3x^2y}{x^4 + 2y^2}$

(a) is equal to 1

(b) is equal to 0

(c) does not exist

(d) is equal to $\dfrac{3}{2}$

(e) is equal to 3

10. The parametric curve

$$x = 2\cos^2 t, \quad y = 1 - \sin t$$

is represented by the rectangular equation

(a) $2x = 2y - y^2,\ x \geq 2$

(b) $x = 4y - 2y^2,\ 0 \leq x \leq 2$

(c) $x = 2y - y^2,\ 0 \leq x \leq 2$

(d) $x = 1 - 4y + 2y^2,\ 0 \leq x \leq 2$

(e) $2x = y - y^2,\ x \geq 0$

11. $\int_{-1}^{1} \int_{0}^{\sqrt{1-x^2}} \int_{0}^{\sqrt{1-x^2-y^2}} (8x^2 + 8y^2 + 8z^2)^{2/3}\, dz\, dy\, dx =$

(a) $\dfrac{\pi}{13}$

(b) $\dfrac{\pi}{6}$

(c) $\dfrac{5\pi}{13}$

(d) $\dfrac{12\pi}{13}$

(e) $\dfrac{3\pi}{8}$

12. The graph of the polar equation $r = 3\cos(6\theta)$ is

(a) a rose with 12 leaves (petals)

(b) a cardioid

(c) a rose with 3 leaves (petals)

(d) a rose with 6 leaves (petals)

(e) a circle

13. The average value of $f(x, y) = 4e^{y^2}$ over the plane region R bounded by the curves $y = 2x$, $y = 4$, and $x = 0$ is equal to

(a) $\dfrac{e^{16} - 1}{2}$

(b) $\dfrac{e^{16} - 1}{16}$

(c) $\dfrac{e^{16} - 1}{8}$

(d) $\dfrac{e^{16} - 1}{4}$

(e) $e^{16} - 1$

14. Let $f(x, y, z) = \ln(x + 2y + 3z)$, $P(1, 1, 1)$, $Q(4, 3, 2)$. The directional derivative of f at P in the direction of the vector $\overrightarrow{PQ}$ is equal to

(a) $\dfrac{2}{\sqrt{14}}$

(b) $\dfrac{7}{6\sqrt{14}}$

(c) $\dfrac{1}{3\sqrt{14}}$

(d) $\dfrac{13}{4\sqrt{14}}$

(e) $\dfrac{5}{3\sqrt{14}}$

15. The set of all points (x, y, z) that are equidistant from the points $(1, 0, 2)$ and $(2, 0, 1)$ represents

(a) a plane

(b) an elliptic hyperboloid

(c) a sphere

(d) an elliptic paraboloid

(e) a line

16. The area of the region bounded by the polar curve $r = \cos\theta$ for $\frac{\pi}{6} \leq \theta \leq \frac{\pi}{3}$ is equal to

(a) $\frac{\pi}{12}$

(b) $\frac{\pi}{8}$

(c) $\frac{\pi}{3}$

(d) $\frac{\pi}{24}$

(e) $\frac{\pi}{6}$

17. If M and m are respectively the maximum and minimum values of $f(x, y) = xy + x + y + 1$ subject to the constraint $xy = 4$, then $M + m =$

(a) 8

(b) 12

(c) 10

(d) 14

(e) 6

18. The volume of the parallelepiped with adjacent edges $\vec{a} = \langle 1, 2, 3\rangle$, $\vec{b} = \langle -1, 1, 2\rangle$, $\vec{c} = \langle 1, -1, 1\rangle$ is equal to

(a) 8

(b) 5

(c) 12

(d) 10

(e) 9

19. If the angle between the two vectors $\vec{u} = \langle 1, 1, -1\rangle$ and $\vec{v} = \langle 2, x, 0\rangle$ is $\frac{\pi}{3}$, then $x^2 + 16x =$

(a) -4

(b) -2

(c) 8

(d) 0

(e) 6

20. $\int_0^2 \int_{y^2}^4 f(x,y)\,dx\,dy =$

(a) $\int_0^4 \int_0^x f(x,y)\,dy\,dx$

(b) $\int_0^2 \int_0^{\sqrt{x}} f(x,y)\,dy\,dx$

(c) $\int_0^2 \int_0^{x^2} f(x,y)\,dy\,dx$

(d) $\int_{y^2}^4 \int_0^2 f(x,y)\,dy\,dx$

(e) $\int_0^4 \int_0^{\sqrt{x}} f(x,y)\,dy\,dx$

21. Let $w = xyz$, $x = s^2 + t^2$, $y = s^2 - t^2$, $z = st$. The value of $\dfrac{\partial w}{\partial t}$ when $s = 2$ and $t = 1$ is equal to

(a) 80

(b) -72

(c) 22

(d) 63

(e) -42

Answer Key

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Q	Ans	Brief explanation
1	**(e)**	Inner gives $4x - \frac{8}{3}$; outer gives $2 - \frac{8}{3} = -\frac{2}{3}$.
2	**(c)**	Right half-disk, z from r^2 to 9, integrand $r^2 \sin\theta$.
3	**(d)**	Polar: $\int_{\pi/2}^{3\pi/2} \int_0^2 e^{-r^2/2} r \, dr \, d\theta = (1 - e^{-2})\pi$.
4	**(e)**	Critical points $(1,1), (-1,-1)$, both with $D = -4 < 0$: two saddle points.
5	**(a)**	Plane $2x - 2y + z + 1 = 0$, so $a + b + c = 2 - 2 + 1 = 1$.
6	**(d)**	Direction $\vec{u} \times \vec{v} = \langle -1, -1, 3 \rangle$: $x = 4 - t,\ y = 3 - t,\ z = 2 + 3t$.
7	**(d)**	Base: triangle $x + y \le 1$, height $3(1 - x - y)$: $V = \frac{1}{2}$.
8	**(d)**	Disk $r \le \sqrt{2}$, height $4 - 2r^2$: $V = 2\pi \cdot 2 = 4\pi$.
9	**(c)**	Path $y = 0$ gives 0, path $y = x^2$ gives 1: limit does not exist.
10	**(b)**	$x = 2(1 - (1 - y)^2) = 4y - 2y^2$, with $0 \le x \le 2$.
11	**(d)**	Spherical: $\int_0^{\pi} \int_0^{\pi/2} \int_0^1 4\rho^{10/3} \sin\phi \, d\rho \, d\phi \, d\theta = \frac{12\pi}{13}$.
12	**(a)**	Rose with $n = 6$ even, so $2n = 12$ petals.
13	**(d)**	$\iint_R f \, dA = e^{16} - 1$, area $= 4$, average $= \frac{e^{16}-1}{4}$.
14	**(e)**	$\nabla f(P) = \frac{1}{6}\langle 1, 2, 3 \rangle$, $\mathbf{u} = \frac{\langle 3,2,1 \rangle}{\sqrt{14}}$: $\frac{5}{3\sqrt{14}}$.
15	**(a)**	Equal distances give $x - z = 0$, a plane.
16	**(d)**	$A = \frac{1}{4} \int (1 + \cos 2\theta) \, d\theta = \frac{1}{4} \cdot \frac{\pi}{6} = \frac{\pi}{24}$.
17	**(c)**	Lagrange: $y = x$, points $(2,2), (-2,-2)$: $M = 9,\ m = 1$, so 10.
18	**(e)**	Triple product determinant $= 9$, so $V = 9$.
19	**(a)**	$2 + x = \frac{1}{2}\sqrt{3}\sqrt{4 + x^2}$, squaring gives $x^2 + 16x = -4$.
20	**(e)**	Region $0 \le x \le 4$, $0 \le y \le \sqrt{x}$.
21	**(c)**	$w_x x_t + w_y y_t + w_z z_t = 12 - 20 + 30 = 22$.